Seed robustness of the gate-unfreeze QFT

100 random-initialisation seeds × three unfreeze orderings, top-20% training objective, DIV2K-8q

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Question

Does the gate-unfreeze QFT endpoint depend on the random draw? We retrain the `QFTBasis(8,8)` operator from **Haar-random** init under the block-growth (`bg`), left→right (`lr`) and right→left (`rl`) unfreeze orderings, for 100 seeds each. Every seed s reseeds **everything trainable** — the Haar gate init, the 50-image training-batch subsample (from the fixed 500-image pool), and the training RNG — while the held-out **test set is held fixed** (canonical seed 42) so the endpoint-PSNR spread reflects the trained operator, not the test draw. Training minimises the top-**20%**-coefficient MSE (the headline elsewhere is top-10%); test PSNR is scored at four keep ratios ρ.

The random initialisations are genuinely different

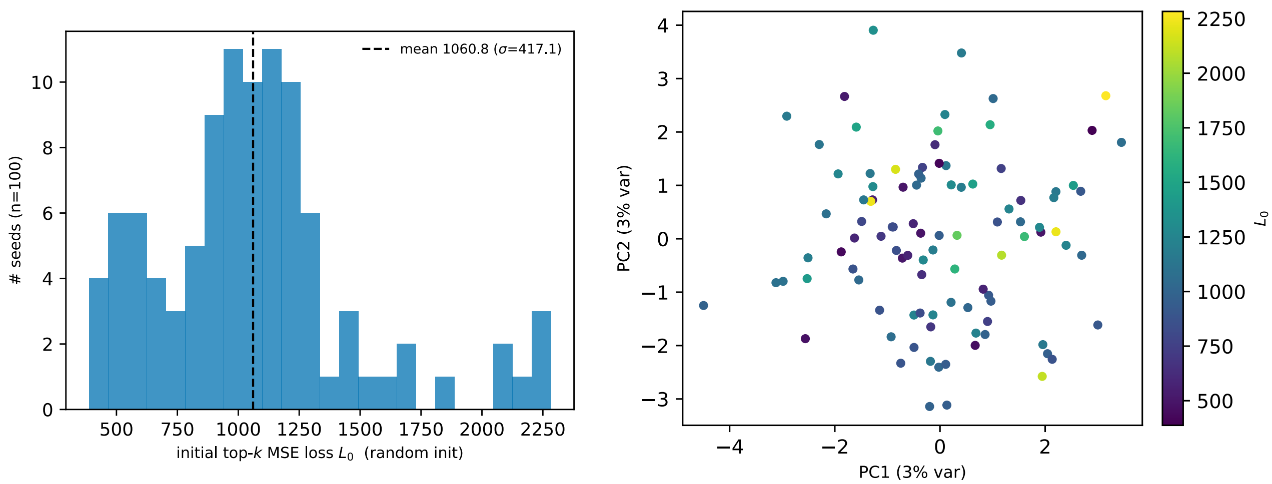

Figure 1: **(a)** the untrained random operator's initial top-k MSE loss L_0 on a common fixed batch, per seed — it spans 387.3–2283.1 (mean 1060.8, $\sigma = 417.1$), so the seeds start from genuinely different points. **(b)** a 2-D PCA of the 100 init parameter vectors; the top two components explain only 2.8% + 2.6% of the variance, i.e. the inits are spread across many near-orthogonal directions rather than clustered.

Endpoint variance

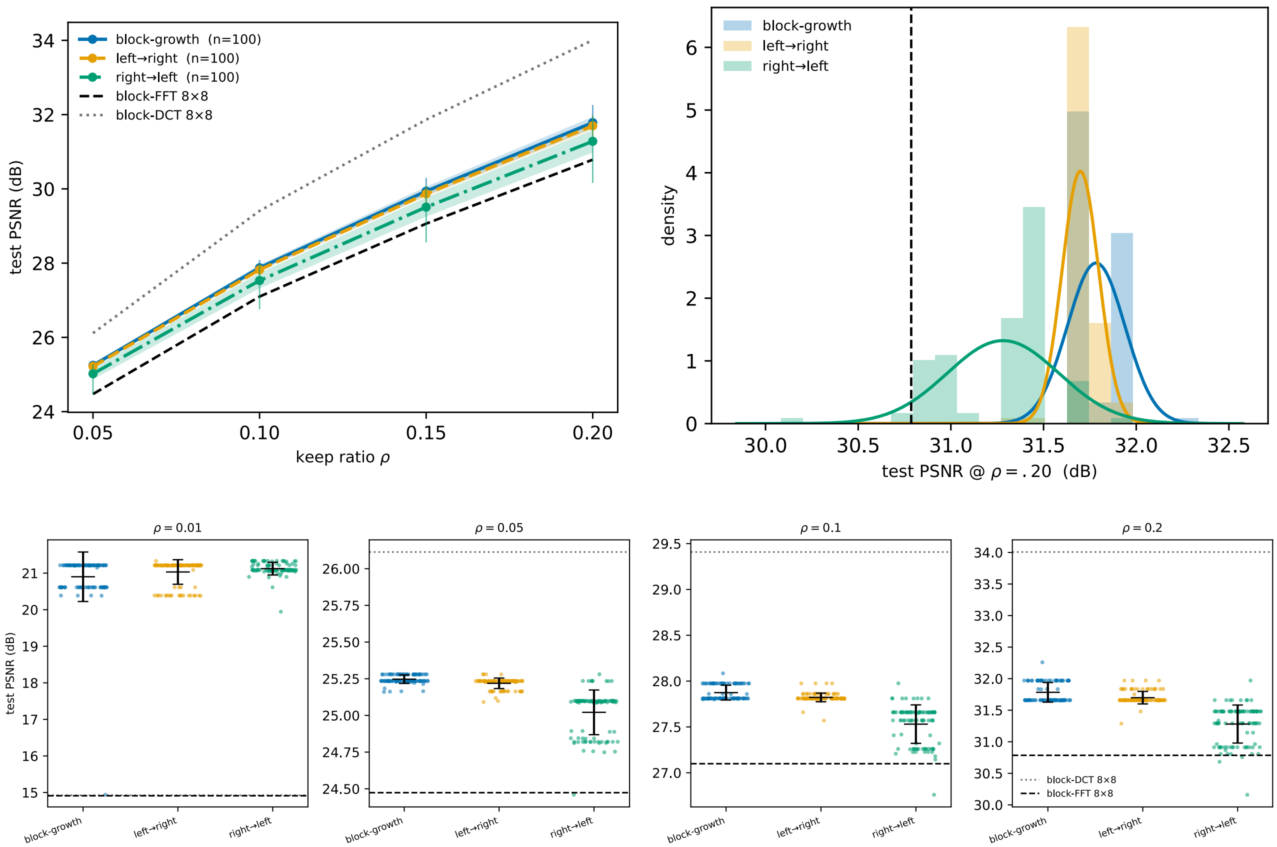

Figure 2: Per-ordering test PSNR across 100 seeds. **Top: (a)** mean $\pm\sigma$ band (+ min– max whiskers) vs ρ, against block-FFT 8×8 (dashed) and block-DCT 8×8 (dotted); and **(c)** the endpoint distribution at ρ{=}.20 (histogram + fitted normal). **Bottom: (b)** per-seed scatter at four keep ratios ($\rho = 0.01, 0.05, 0.10, 0.20$) — the trained bases beat block-FFT at every rate and exceed both classical references by ~ 6 dB at ρ{=}.01, while block-DCT overtakes them for $\rho \geq .05$.

ordering	ρ{=}.05	ρ{=}.10	ρ{=}.15	ρ{=}.20	**min@.20**
block-growth	**25.25±0.03**	**27.87±0.08**	**29.94±0.12**	**31.78±0.16**	31.65
left→right	25.22±0.04	27.82±0.05	29.87±0.07	31.7±0.1	31.29
right→left	25.02±0.15	27.53±0.21	29.51±0.26	31.28±0.3	30.16
block-FFT 8×8	24.47	27.1	29.06	30.79	—
block-DCT 8×8	26.11	29.41	31.86	34.01	—

mean $\pm\sigma$ test PSNR (dB), $n = 100$ random-init seeds per ordering; **min@.20** is the single worst seed at ρ{=}.20.

Reading

Despite starting from 100 genuinely different random inits (previous section), each ordering's endpoint sits in a narrow band: at ρ{=}.20 the per-seed standard deviation is block-growth 0.16, left→right 0.1, right→left 0.3 dB. The **release order**, not the random draw, is the dominant factor: block-growth and left→right are tight and sit highest, while right→left is ~ 2× wider and lowest — the dynamics figure below explains why (rl starts from a far worse loss and its seeds stay spread until the final unfreeze stages).

Against the classical block-FFT 8×8 reference (30.79 dB @ρ{=}.20): block-growth and left→right clear it for **every** seed (worst seed 31.65 / 31.29 dB; margins +0.86 / +0.5 dB). right→left clears it for 97 of 100 seeds; 3 low-tail seeds dip below by at most 0.63 dB. Overall 297 of 300 runs (99%) clear block-FFT. So the gate-unfreeze QFT is **essentially always** above block-FFT — unconditionally for bg/lr, and for 97% of the highest-variance rl ordering — while block-DCT 8×8 (34.01 dB) remains the strongest classical transform here, ahead of the QFT family.

The per-seed endpoints are **not** Gaussian: a Shapiro– Wilk test rejects normality for all three orderings ($p < 10^{-7}$), because seeds settle into a few discrete attractor basins of the very-flat top-k MSE valley rather than scattering smoothly. The fitted normal in the endpoint panel is a visual summary of location and spread, not a claim of Gaussianity.

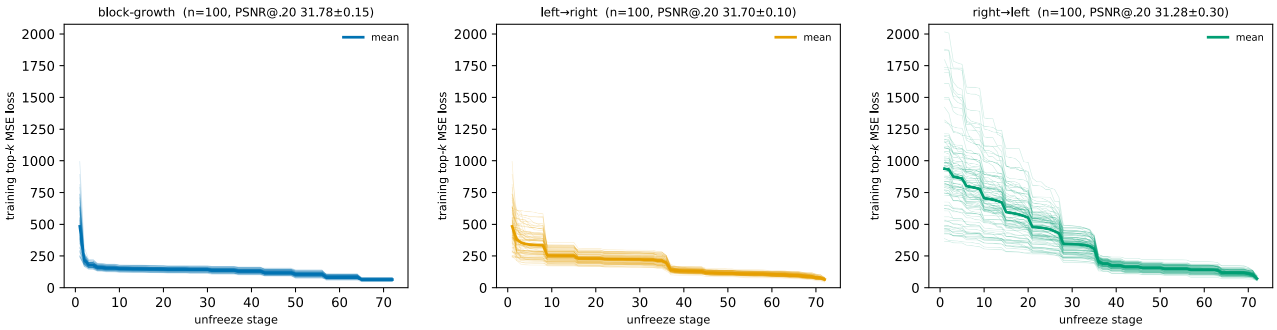

Figure 3: Per-seed training dynamics: every seed's top-k MSE staircase (one per unfreeze stage), one panel per ordering, on a shared y-range. Different Haar starts descend along different paths: block-growth **(a)** collapses fast and stays tight, whereas right→left **(c)** starts far higher and its seeds remain spread until the final stages — visually accounting for the endpoint variances above. left→right **(b)** is intermediate.

Emergent block structure of the trained QFT

We close by reading the **canonical published** trained QFT (`QFTBasis(8,8)`, seed 98) directly, making the paper's central structural claim concrete (sec5.3): the trained full-image QFT factors itself into a block code with no block prior. Of its 16 Hadamard-role gates (8 per dimension), 4 per dimension collapse to non-mixing **Pauli-Z/X** gates (the off-diagonal/diagonal modulus drops to $\approx 0.3\%$) — classical block indices — while 53 of the 56 controlled-phase gates stay active, so the intra-block transform stays rich. The 4 still-mixing row qubits and 4 column qubits leave an effective 16×16-pixel block.

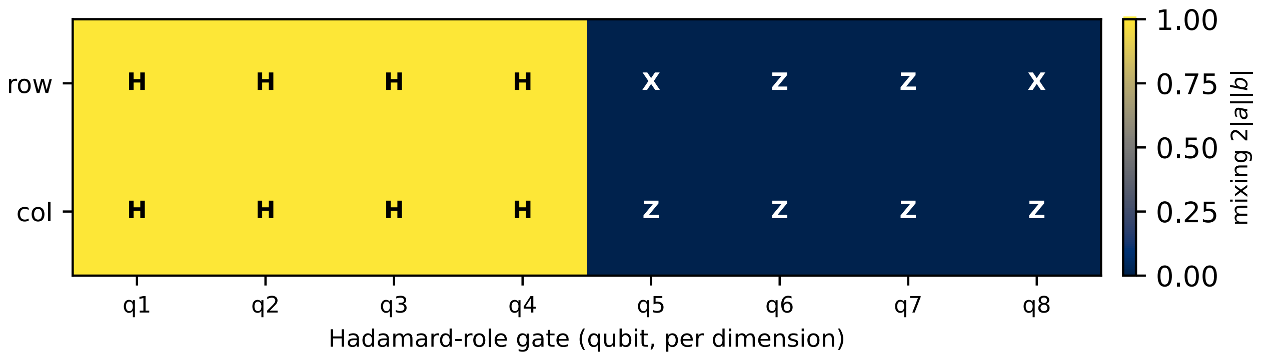

Figure 4: Hadamard-role gates of the trained QFT, one cell per qubit and dimension, shaded by the mixing score $2|a||b|$ ($1 =$ Hadamard, $0 =$ frozen) and labelled by type. Four of the eight Hadamards per dimension keep mixing frequencies; the other four freeze to Pauli-Z (or X) and act as classical block indices.

The factorization is a statement about the trained operator itself. The transform is separable, $Y = WXW^{\top}$, so its block structure lives in the single $N \times N$ ($N = 2^8 = 256$) one-dimensional factor W. Acting on an input row x it returns the coefficient row

$$y = Wx, \quad y_i = \sum_{j=0}^{N-1} W_{ij} x_j,$$

where j indexes the input pixel and i the output coefficient; the j-th column of W is the operator's response to a single input pixel, We_j. Grouping the 256 indices into sixteen contiguous 16-pixel blocks $B_p = \{16p, ..., 16p + 15\}$, the trained W is **block-diagonal up to a block permutation** σ: $j \in B_q$ implies $W_{ij} = 0$ for all $i \notin B_{\sigma(q)}$, i.e.

$$P^\top W = W^{(0)} \oplus \cdots \oplus W^{(15)}, \quad W^{(p)} \in \mathbb{C}^{16\times 16},$$

a direct sum of sixteen 16×16 within-block transforms $W^{(p)}$ (the permutation σ, and the off-diagonal placement of the blocks, come from the QFT bit-reversal and the frozen-X gates). Each input pixel therefore drives only the 16 output coefficients of its own block. Figure 5 shows this two ways: the full matrix $|W|$ (left) and three of its columns We_j (right). The untrained global QFT, by contrast, has $|W_{ij}| = 1/\sqrt{N}$ for every i, j, so a single pixel spreads across all 256 coefficients — no blocks.

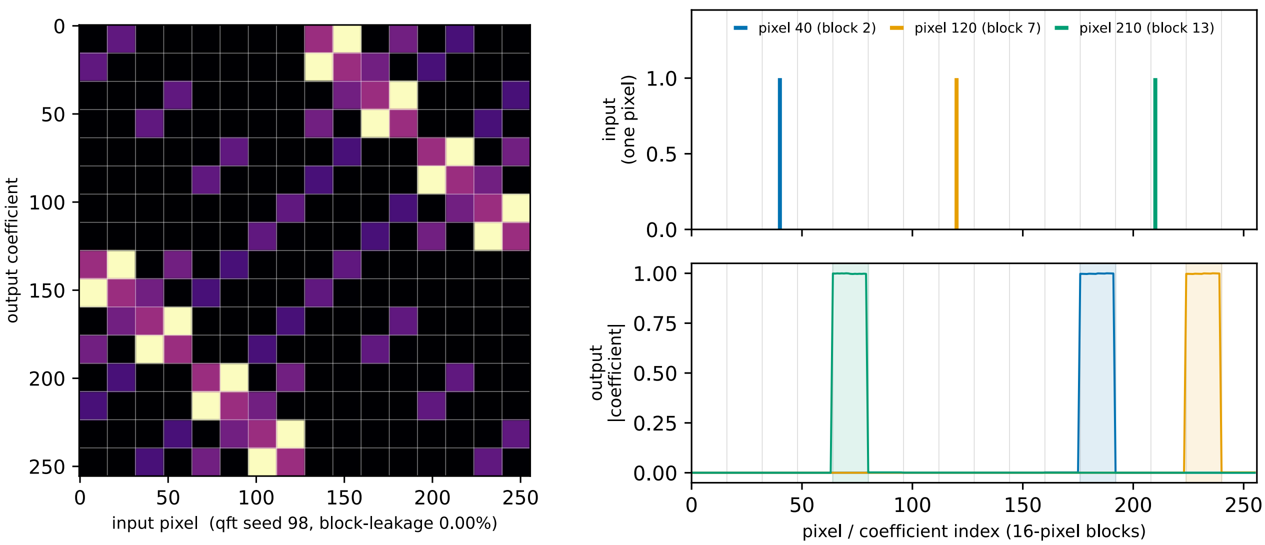

Figure 5: **Left:** $\log|W|$ of the trained QFT's 1-D operator factor, block-diagonal at the emergent 16-pixel block (off-block energy below 0.01%). **Right:** the same operator read one input at a time — a single input pixel (top) produces output (bottom) confined to a single 16-coefficient block ($> 99.99\%$ of its energy), zero elsewhere. The output block is permuted from the input one by the QFT bit-reversal and the frozen-X gates; a global transform would instead spread each pixel across all 256 coefficients. The heatmap is these per-pixel responses stacked side by side.

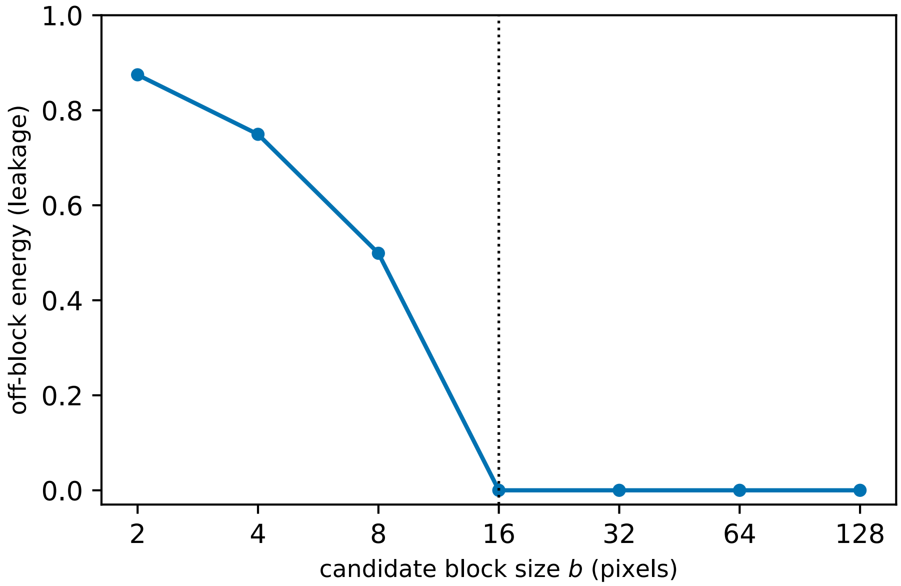

Figure 6: Off-block energy vs candidate block size — the knee at $b = 16$ (dotted) shows the operator is block-structured at that specific scale and not below; the untrained closed-form QFT (not shown) stays high at every scale.

The same factorization is visible directly in frequency space.

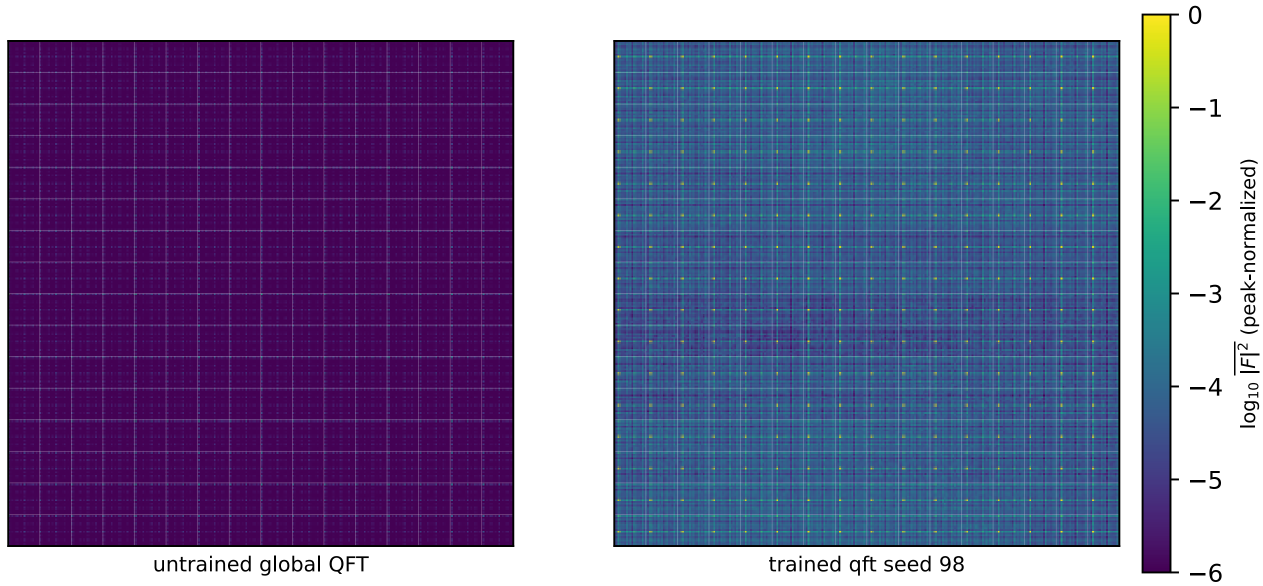

Figure 7: Mean test-set power spectrum (peak-normalized $\log_{10} \overline{|F|^2}$, averaged over the 50 DIV2K test images) of the untrained **global QFT** (left) and the **trained QFT** (right). The global transform packs energy into a single low-frequency lobe; the trained transform tiles into a 16×16 grid of repeated sub-spectra — the block-periodic signature of a block transform, the same structure the 8×8 block references produce in Fig. 4 of the main text.